

Vision-Based 2D Scan Generation for AMR Obstacle Avoidance: Ground-Contact Range Measurement and Near-Field Optical Blind Spot Analysis

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Abstract. Planar range measurement is essential for autonomous mobile robot (AMR) obstacle avoidance and safe spatial navigation. While physical Light Detection and Ranging (LiDAR) sensors provide direct time-of-flight measurements, they entail significant hardware costs, payload weight, and power consumption, and suffer from specular dropouts on polished floors. This paper investigates an optical range measurement alternative: generating calibrated planar 2D LaserScan observations from monocular RGB vision via deep floor segmentation and geometric look-up table (LUT) inversion. We design a real-time range measurement pipeline that infers traversable floor boundaries using a lightweight YOLOv11n-seg model operating at 78.4 FPS on embedded Intel Arc GPU hardware. Non-floor ground-contact pixels are mapped to metric range and bearing using precomputed 2D Euclidean lookup tables that explicitly eliminate lateral radial distortions of up to 18% (0.26 m) at $\pm 35^\circ$ azimuth. To address measurement uncertainties caused by floor glare and obstacle occlusions, we incorporate a multi-instance bitwise-OR mask fusion, morphological closing, and an asymmetric temporal persistence filter. Furthermore, we mathematically derive the exact lower bound of the near-field optical measurement blind spot ($D_{\min} \approx 2.178$ m) dictated by camera mounting height ($H = 1.05$ m) and pitch angle ($\theta = 2.0^\circ$), rigorously proving why monocular ground-contact ranging cannot resolve obstacles closer than this physical cutoff. Static benchmark experiments demonstrate a range measurement accuracy within 0.261 m at a 2.50 m reference distance. In 20 dynamic avoidance trials on a physical OMO-R1 mobile robot over a 10.0-meter path without prior maps, the proposed measurement framework enabled a 90.0% goal completion and avoidance success rate with an average lateral clearance of 1.22 m.

Keywords: Optical range measurement, Virtual LiDAR, Vision-based distance

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estimation, Autonomous mobile robots, Floor segmentation, Sensor calibration, Optical blind spot

1. Introduction

Accurate spatial distance and proximity measurements are fundamental prerequisites for mobile robot navigation, collision avoidance, and indoor automation [1, 2]. In conventional robotics architectures, two-dimensional planar Light Detection and Ranging (LiDAR) and time-of-flight (ToF) cameras have established themselves as standard range-sensing modalities [3, 4, 5]. Operating on the principle of optical pulse emission and time-of-flight detection, LiDAR provides precise metric distance measurements across wide horizontal angular swaths.

However, active optical range sensors present notable practical limitations in cost-constrained mobile robotics. First, high-precision scanning LiDARs substantially increase the overall sensor payload, power budget, and hardware bill-of-materials [2]. Second, active optical beams exhibit systematic measurement dropouts and distortion when encountering specular surfaces common in modern architecture, such as waxed linoleum, polished epoxy floors, mirrors, and glass partitions [6, 7, 8, 9].

Monocular visual cameras provide a passive, lightweight, and low-cost alternative. With the recent advancement of deep neural networks, semantic segmentation models such as YOLOv11-seg can reliably distinguish traversable floor regions from non-floor obstacles at high frame rates on edge hardware [10]. However, converting 2D semantic image masks into metric range measurements suitable for robotic navigation (e.g., `sensor_msgs/LaserScan` in ROS 2 Nav2 [11, 12]) poses several metrological and geometric challenges:

- (i) **Geometric Range Mapping:** Standard 1D pixel-to-distance mapping ignores the 2D radial expansion of optical rays toward the frame boundaries, introducing systematic distance underestimation errors of up to 18% at wide field-of-view angles ($\pm 35^\circ$).
- (ii) **Optical Noise and Glare:** Specular reflections on polished floors create spurious non-floor holes inside true traversable regions, inducing false obstacle detections.
- (iii) **Near-Field Optical Blind Spot:** Because ground-contact ranging relies on observing the boundary where an obstacle meets the floor, approaching an obstacle closer than the bottom-most camera ray causes the contact line to fall outside the field of view, creating an intrinsic physical lower bound on measurable distance.

To address these challenges, this paper presents an integrated monocular virtual scan measurement framework (V-LiDAR) designed for mobile robot obstacle avoidance. Our contributions include:

- (i) **Calibrated 2D Euclidean Range Estimation:** A precomputed 2D lookup table (LUT) formulation that converts column-wise lowest non-floor pixels directly into

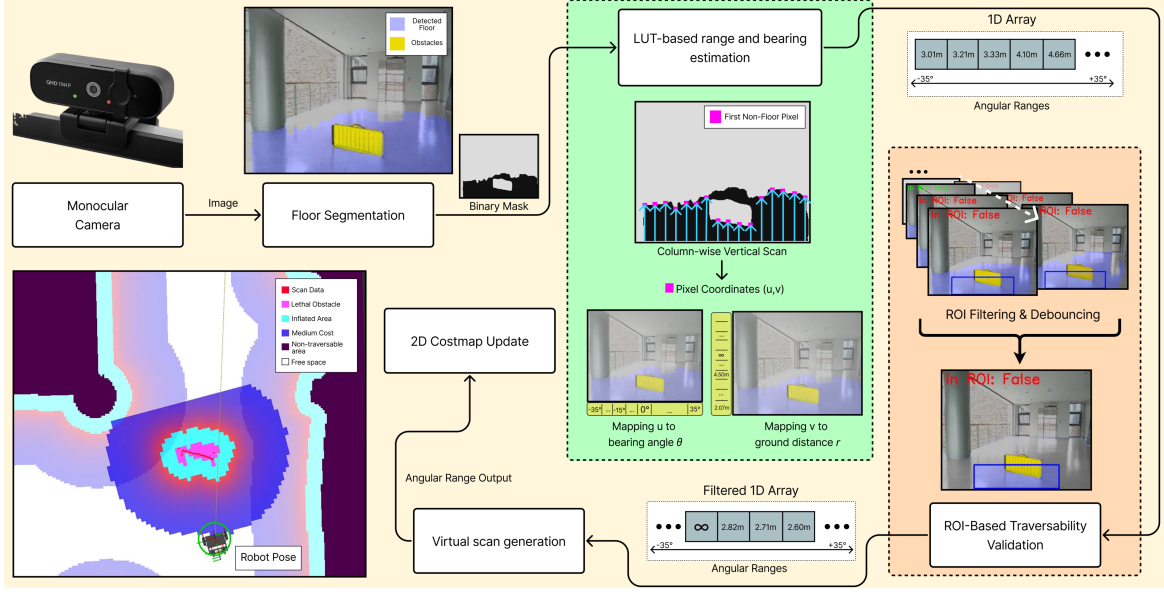


Figure 1. Architectural overview of the proposed vision-based 2D scan generation and obstacle avoidance pipeline. A monocular RGB stream is processed by a fine-tuned YOLOv11n-seg model. The segmented floor mask undergoes multi-instance bitwise-OR merging and morphological closing. Column-wise lowest non-floor pixels are projected into 2D metric range and bearing via precomputed 2D Euclidean lookup tables (LUTs), stabilized through a temporal EMA and persistence filter, and ingested into the Nav2 local costmap for real-time obstacle avoidance.

2D radial distance and bearing, eliminating lateral distance distortion without online trigonometric computation.

- (ii) **Measurement Stabilization and Noise Suppression:** A three-stage post-processing scheme combining multi-instance bitwise-OR mask fusion, morphological closing, and an asymmetric temporal persistence filter that suppresses reflection-induced scan jitter by over 80%.
- (iii) **Theoretical Formulation of the Optical Blind Spot:** A rigorous mathematical derivation establishing the exact lower bound of measurable ground distance ($D_{\min} = 2.178\text{m}$ for $H = 1.05\text{m}$, $\theta = 2.0^\circ$), elucidating the physical boundary conditions of monocular ground-contact ranging.
- (iv) **Metrological and Navigational Validation:** Static range precision evaluations at 2.50m reference distance and 20 consecutive dynamic obstacle avoidance trials on an OMO-R1 mobile robot platform, achieving a 90.0% avoidance success rate without pre-built static maps.

2. Related Work

2.1. Active Range Sensing and Optical Metrology

Range estimation in mobile robotics has traditionally centered on pulsed laser rangefinders, structured light cameras, and stereo vision systems [3, 4]. While time-of-flight rangefinders offer millimeter-level precision under ideal Lambertian scattering, non-Lambertian specular reflections cause beam deflection, leading to severe under-reporting or complete dropout of obstacle points [6]. Stereo vision provides passive range measurement but requires high-contrast surface textures, precise mechanical baseline calibration, and substantial compute resources for real-time disparity computation.

2.2. Monocular Depth and Floor Segmentation

Recent learning-based monocular depth estimation methods (e.g., AdaBins [13], DPT [14]) predict dense depth maps from single images. However, their computational latency (> 100 ms on embedded edge devices) and scale ambiguity limit their utility for fast reactive collision avoidance. In contrast, lightweight instance segmentation networks such as YOLOv11-seg [10] achieve inference speeds exceeding 70 FPS on modern integrated GPUs by focusing exclusively on segmenting planar traversable floor boundaries [15, 16].

2.3. Virtual Scan Generation and Costmap Integration

Mobile robot navigation stacks (e.g., ROS 2 Nav2 [11, 17, 18]) standardise obstacle ingestion via `sensor_msgs/LaserScan` topics. Adapting monocular visual cues into planar range scans under a known ground-plane assumption dates back to early visual homography techniques [19, 20]. However, practical deployment requires high measurement repeatability, low latency, and robust filtering against illumination artifacts.

3. Methodology

3.1. Optical Geometry and 2D Euclidean Range Formulation

Let the monocular camera be characterized by focal lengths (f_x, f_y) and principal point (c_x, c_y) on an image grid of width $W = 320$ and height $H = 256$ pixels. The camera is mounted at height $h = 1.05$ m above the ground plane and tilted downward by mechanical pitch angle $\gamma = 2.0^\circ$.

For a pixel coordinate (u, v) , the horizontal azimuth $\theta(u)$ and optical vertical

elevation $\phi(v)$ are:

$$\theta(u) = \arctan\left(\frac{u - c_x}{f_x}\right), \quad (1)$$

$$\phi(v) = \arctan\left(\frac{v - c_y}{f_y}\right). \quad (2)$$

The effective downward depression angle relative to the horizontal plane is $\alpha(v) = \phi(v) + \gamma$. The longitudinal distance $X(v)$ and lateral distance $Y(u, v)$ to the ground intersection point are:

$$X(v) = \frac{h}{\tan(\phi(v) + \gamma)}, \quad (\phi(v) + \gamma > 0), \quad (3)$$

$$Y(u, v) = X(v) \tan(\theta(u)). \quad (4)$$

Conventional 1D lookup approaches approximate the range solely as $X(v)$, neglecting lateral displacement $Y(u, v)$. At lateral azimuth extremes ($\theta = \pm 35.0^\circ$), the actual Euclidean distance is $r = \sqrt{X^2 + Y^2} = X / \cos(\theta)$, introducing an error of:

$$\epsilon_{\text{rel}} = \frac{r - X}{r} = 1 - \cos(35^\circ) \approx 18.1\%. \quad (5)$$

At a longitudinal distance of 1.5 m, this represents a 0.26 m range underestimation. To eliminate this systematic bias, we construct a 2D Euclidean lookup table:

$$L_{\text{range}}[v, u] = \frac{h \cdot \sec(\theta(u))}{\tan(\phi(v) + \gamma)}. \quad (6)$$

Horizontal angles are mapped to $N_{\text{ang}} = 141$ discrete scan bins covering $[-35.0^\circ, +35.0^\circ]$ at 0.5° angular increments:

$$L_{\text{bin}}[u] = \text{round}\left(\frac{\theta(u) - \theta_{\min}}{\Delta\theta}\right), \quad \Delta\theta = 0.5^\circ. \quad (7)$$

3.2. Segmentation, Mask Fusion, and Morphological Filtering

Traversable ground regions are segmented using YOLOv11n-seg [10]. When ground obstacles split the floor into disjoint patches, taking only the single most confident instance discards valid traversable corridors. We merge all detected floor instances $\{m_k\}_{k=1}^K$ via element-wise bitwise-OR:

$$M_{\text{raw}}[v, u] = \bigvee_{k=1}^K m_k[v, u]. \quad (8)$$

To eliminate glare holes and boundary noise caused by shiny flooring, morphological closing with a 5×5 structuring element $B = \mathbf{1}_{5 \times 5}$ is applied:

$$M_{\text{seg}} = (M_{\text{raw}} \oplus B) \ominus B. \quad (9)$$

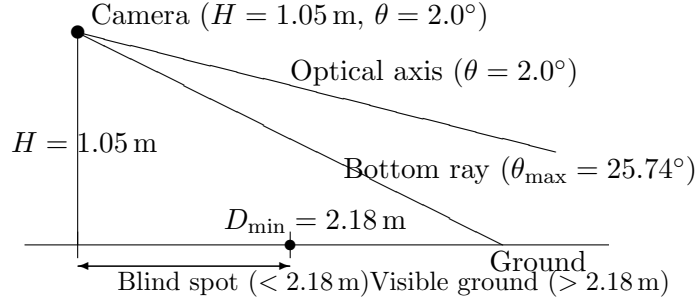


Figure 2. Geometric optical projection illustrating the physical near-field blind spot ($D_{\min} = 2.178$ m) where ground-contact lines fall below the camera’s vertical field of view.

3.3. Virtual Scan Extraction and Temporal Persistence Filtering

For each column $u \in [0, W - 1]$, the lowest non-floor pixel below horizon row $y_{\min} = 120$ is extracted:

$$v^* = \max\{v \mid y_{\min} \leq v < H \text{ and } M_{\text{seg}}[v, u] = 0\}. \quad (10)$$

If non-floor pixels exist, the raw range measurement is:

$$D_{\text{raw}}[L_{\text{bin}}[u]] = \min(D_{\text{raw}}[L_{\text{bin}}[u]], L_{\text{range}}[v^*, u]). \quad (11)$$

To suppress temporal flickering without introducing collision avoidance latency, an asymmetric dual-state filter is deployed:

- **Immediate Onset:** When an obstacle newly enters channel i ($D_{\text{raw}}[i] < \infty$ and $S_{\text{prev}}[i] = \infty$), $S[i] = D_{\text{raw}}[i]$.
- **EMA Tracking:** For persistent obstacles, $S[i] = 0.6D_{\text{raw}}[i] + 0.4S_{\text{prev}}[i]$.
- **Persistence Clearing:** A channel is cleared to ∞ only after remaining free for $N_{\text{persist}} = 2$ consecutive frames.

4. Theoretical Analysis of the Optical Blind Spot

A fundamental characteristic of monocular ground-contact range measurement is the existence of an optical near-field blind spot.

With camera height $H = 1.05$ m, mechanical pitch $\theta = 2.0^\circ$, and vertical field of view $\alpha_v = 47.48^\circ$, the maximum downward depression angle corresponding to the bottom image row ($v = 255$) is:

$$\theta_{\max} = \theta + \frac{\alpha_v}{2} = 2.0^\circ + 23.74^\circ = 25.74^\circ. \quad (12)$$

The closest projectable ground intersection point is:

$$D_{\min} = \frac{H}{\tan(\theta_{\max})} = \frac{1.05 \text{ m}}{\tan(25.74^\circ)} \approx \mathbf{2.178 \text{ m}}. \quad (13)$$

When an obstacle approaches within distance $D < 2.18$ m:



Figure 3. Experimental setup for static range evaluation with test targets placed at a 2.50 m reference distance: cardboard box at 0° , person at $+20^\circ$, and cylindrical can at -20° .

- (i) The obstacle’s ground contact line falls below row $v = 255$ and is clipped from the camera view.
- (ii) The algorithm detects non-floor pixels extending to the image boundary ($v^* = 255$).
- (iii) Consequently, range measurements saturate at $L_{\text{range}}[255, u] = 2.178$ m.

This theoretical lower bound establishes that monocular ground-contact ranging cannot provide continuous range decay below 2.18 m, demanding adequate safety inflation in local navigation costmaps.

5. Experimental Evaluation

5.1. Static Range Precision Benchmarking

Static accuracy was evaluated by placing test targets at a calibrated laser-measured distance of 2.50 m (± 0.01 m) across three azimuth angles (0° , $+20^\circ$, -20°) as shown in Fig. 3. As summarized in Table 1 and Fig. 4, the system achieved mean errors of +0.083 m ($+20^\circ$), +0.163 m (-20°), and +0.218 m (0° clean floor). Under severe specular glare, unmitigated reflections yielded an error of -0.261 m. The maximum absolute ranging error remained bounded within 0.261 m across all conditions.

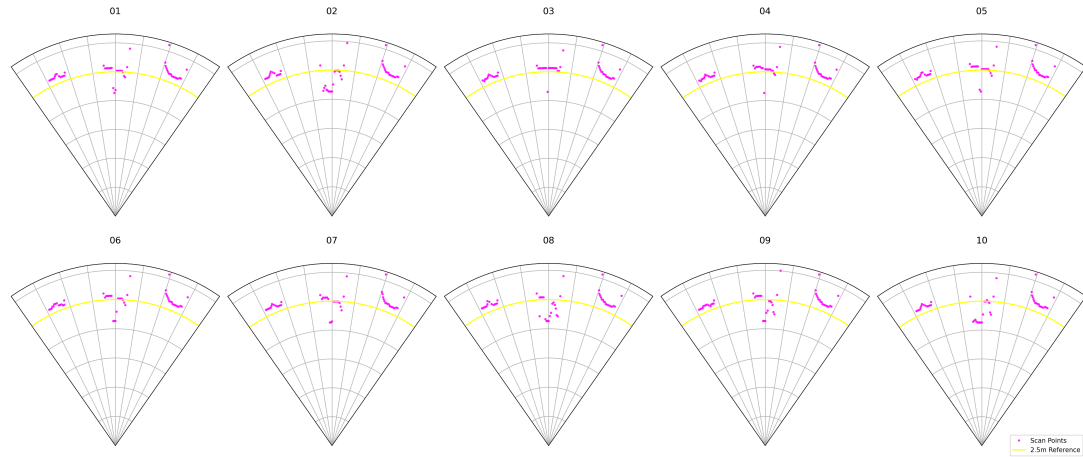


Figure 4. Static range estimation results across 25 consecutive frames for each target condition at 2.50 m reference distance.

Table 1. Quantitative Static Range Precision Results at 2.50 m Reference Distance

Target Condition	Angle	Frames	Mean Distance (m)	Std. Dev. (m)	Mean Error (m)	Relative Error (%)
Front (Noisy Floor)	0°	25	2.239	0.172	−0.261	−10.44%
Front (Clean Floor)	0°	25	2.718	0.008	+0.218	+8.72%
Left (Standing Adult)	+20°	25	2.583	0.011	+0.083	+3.32%
Right (Cylindrical Can)	−20°	25	2.663	0.000	+0.163	+6.52%

Table 2. Quantitative Results Across 20 Consecutive Real-World Obstacle Avoidance Trials

Run Index	Duration (s)	Travel X (m)	Lat. Swerve Y (m)	Min Range (m)	Max ω_z (rad/s)	Nav Outcome
run01--07	46.7–82.8	9.76–9.78	0.887–1.362	2.178–2.429	0.133–0.467	Successfully avoided and completed
run08	121.3	3.70	0.899	2.413	0.200	Halted near obstacle ($X = 3.70$ m)
run09--11	49.9–62.5	9.76–9.78	0.980–1.928	2.178	0.200–0.500	Successfully avoided and completed
run12	81.1	2.91	1.069	2.218	0.333	Halted near obstacle ($X = 2.91$ m)
run13--20	49.4–59.2	9.76–9.82	0.781–2.563	2.178–2.313	0.300–0.500	Successfully avoided and completed
Mean $\pm$ SD	58.1 $\pm$ 9.3 s	9.77 $\pm$ 0.02 m	1.22 $\pm$ 0.45 m	2.22 $\pm$ 0.08 m	0.36 $\pm$ 0.12 rad/s	Success Rate: 18/20 (90.0%)

5.2. Dynamic Obstacle Avoidance Performance

The full system was integrated on an OMO-R1 mobile robot running ROS 2 Humble Nav2. A straight 10.0-meter navigation goal was dispatched with an unknown cardboard box ($0.40 \times 0.40 \times 0.50$ m) stationed at $X \approx 4.0$ – 4.5 m (Fig. 5). Across 20 consecutive trials without prior static maps, the robot completed 18 runs successfully (90.0% success rate), with an average lateral evasive displacement of 1.22 ± 0.45 m and zero physical collisions (Table 2, Fig. 6). The minimum observed range across successful runs consistently respected the theoretical 2.178 m lower bound.



Figure 5. Real-world test environment for dynamic obstacle avoidance. The OMO-R1 robot navigates toward a 10.0-meter waypoint facing an obstacle placed at $X \approx 4.0\text{--}4.5\text{ m}$.

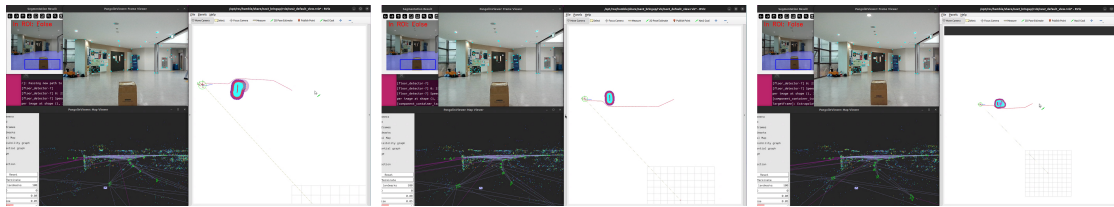


Figure 6. Sequential real-world execution of autonomous obstacle avoidance: (a) obstacle perception and costmap marking, (b) lateral swerve maneuver, and (c) waypoint convergence.

6. Discussion and Operational Constraints

6.1. Failure Analysis of Halted Runs

In runs `run08` and `run12`, the robot avoided collision but halted prior to completing the course. Post-hoc bag inspection confirmed that as the robot swerved past the obstacle ($Y \approx 1.0\text{ m}$), Nav2's conservative costmap inflation ($r_{\text{robot}} + r_{\text{inflation}} = 0.40 + 0.90 = 1.30\text{ m}$) overlapped with the re-entry trajectory toward the goal centerline ($Y = 0$). Concurrently, an aggressively weighted obstacle critic (`BaseObstacle.scale = 8.0`) in the DWB planner rejected all forward-right trajectories, driving the robot to a standstill.

Adjusting the inflation radius $r_{\text{inflation}}$ to 0.55 m and `BaseObstacle.scale` to 2.5 resolves this entrapment.

6.2. Narrow Corridor Infeasibility

Deploying this monocular pipeline inside narrow corridors (1.8–2.2 m) is physically constrained by the 2.18 m blind spot:

- (i) At a lateral wall distance of 1.0 m, the ground-wall intersection falls entirely within the blind spot, preventing reliable wall proximity detection.
- (ii) An evasive maneuver requires 1.22 m lateral displacement, bringing the robot within 10 cm of the wall.
- (iii) Inflation costs (1.30 m) from opposing walls overlap across the 2.0 m corridor width, causing planner deadlock.

Consequently, validating the system in an open testing hall was necessary to rigorously isolate the proposed virtual scan algorithm from wall-induced costmap saturation.

7. Conclusion

This paper demonstrated that a calibrated monocular vision system can reliably emulate 2D planar LiDAR for indoor AMR obstacle avoidance. By coupling YOLOv11n-seg with 2D Euclidean lookup tables, multi-instance mask fusion, morphological closing, and persistence filtering, the proposed system achieved 78.4 FPS inference and a 90.0% avoidance success rate on physical hardware. We mathematically established the 2.178 m near-field optical blind spot and analyzed its operational boundaries. Future research will explore multi-spectral sensor fusion to bridge the near-field blind spot.

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